

Group 3 — Idea 2 (SmartSort — AI-Based Automatic Waste Segregation Robot)

My overall take

This is the one we want you to move forward with. The perception-decision-actuation loop is well thought out, and using a confidence threshold to gate ambiguous items instead of forcing a guess is a genuinely good design decision. That said, I want you to simplify from three categories down to two, and add a proper bin-management system with user notification.

What would make this project feel complete [improvement suggestions]

- Every item gets sorted into one of the two; no reject tray. If confidence is genuinely too low on a specific item, treat that as a model-tuning problem to solve, not something to route around with a third bin.
- Instead of a flap that angles the falling item into a pile, I want an actual arm, two joints is enough, that physically picks the item up after classification and places it into the correct pile. This is a stronger demonstration of actuator control and decision-driven manipulation than a passive gravity-fed flap, and it's more in line with what this course is testing. Whatever you build, it needs to actually work end-to-end with the fill/dump system described below
- When a compartment's fill sensor crosses your full threshold, send a notification to the user, and let them choose one of two options: empty it themselves, or have the system handle it. If the user chooses the system to handle it, there should be a mechanism to dump each pile's trash to another bin and then continue with its process. One way to do this would be stacking two containers on top of each other with a latch system in between, but if you can come up with something better do that.
- Odor control mechanism is only valid if you're building this for a household environment. In that case, there should be a mechanism to check air quality and then release the scent. But again, you may skip this if the robot is meant for more industrial applications.
- You listed ≥85% held-out accuracy for the classifier. I want error rate at or below 5% meaning the system should classify correctly at least 95% of the time. Given how controlled your imaging chamber already is (fixed lighting, background, distance), this should be achievable. The buzzer can double as your error/alert signal when this threshold isn't met or an item can't be confidently sorted.

Please test these before the final presentation

- Run the sorting mechanism through repeated cycles with real test items and confirm it consistently and correctly routes at max 50g objects into the right pile, whichever mechanism you go with.
- Test the full notification → user choice → dump flap sequence end to end, for both the "user empties it" and "system handles it" paths.
- Confirm the dump mechanism reliably clears the bin without jamming across a range of item sizes/weights.

A good final demonstration would look like this
An item is placed, the system classifies it, the arm picks it up and places it in the correct pile. Do this for at least one recyclable and one biodegradable item so both classes are shown. Then simulate a full bin (or actually fill it) to trigger the notification, show both user response paths (manual empty vs. system-handled), and if "let the system handle it" is chosen, show the dump mechanism actually clearing the bin. Also you can show the buzzer works when there is an error and user is notified via a notification that something went wrong.
Keeping the work manageable
Use a small rectangular arena, two agreed colours and one lightweight object shape. Predefined arm poses are completely acceptable. Finish stationary pick-and-place before connecting the arm to autonomous navigation.
One final note
<i>Keep the object set simple and make the behaviour repeatable. Two colours and one object shape are enough. The quality of the pickup and the complete autonomous sequence will matter more than recognizing many different types of trash.</i>